

Interferometers

Michelson Interferometer

- The Michelson interferometer is the best example of what is called an amplitude-splitting interferometer
- It was invented in 1893 by Albert Michelson, to measure a standard meter in units of the wavelength of the red line of the cadmium spectrum
- With an optical interferometer, one can measure distances directly in terms of wavelength of light used, by counting the interference fringes that move when one or the other of two mirrors are moved

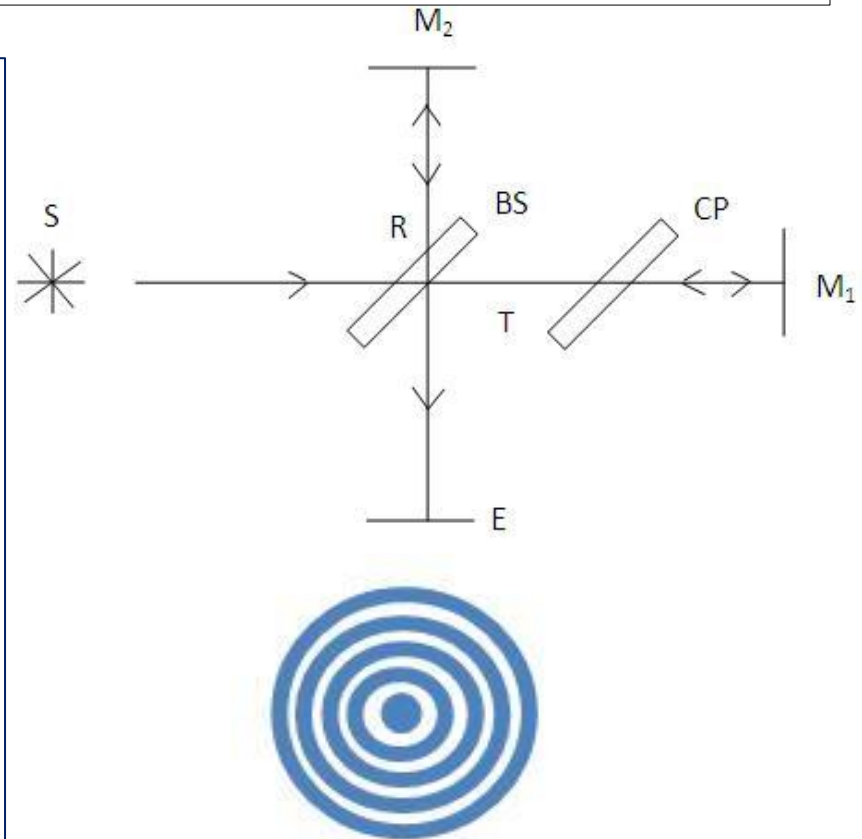
- In the Michelson interferometer, coherent beams are obtained by splitting a beam of light that originates from a single source with a partially reflecting mirror called a beam splitter
- The resulting reflected and transmitted waves are then re-directed by ordinary mirrors to a screen where they superimpose to create fringes. This is known as interference by division of amplitude.

Principle of Michelson's Interferometer

- In Michelson's Interferometer, a beam of light from an extended source is divided into two parts of equal intensities by partial reflection and refraction
- These beams travel in two mutually perpendicular directions and come together after reflection from plane mirrors
- The beam overlap on each other and produce interference fringes

Construction and working

- Light from a monochromatic source S is divided by a beam splitter (BS), which is oriented at an angle 45° to the beam, producing two beams of equal intensity.
- The transmitted beam (T) travels to mirror M_1 and it is reflected back to BS.
- 50% of the returning beam is then reflected by the beam splitter and strikes the screen, E.



The reflected beam (R) travels to mirror M_2 , where it is reflected.

50% of this beam passes straight through beam splitter and reaches the screen.

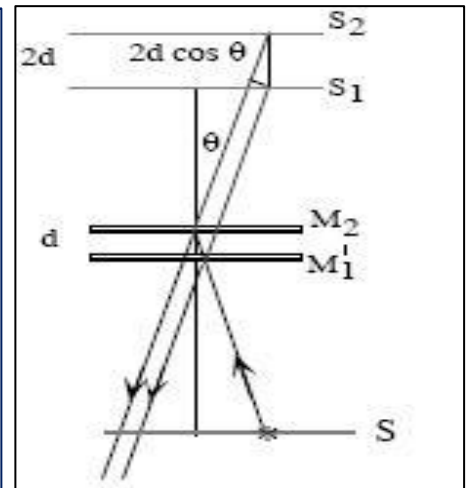
- Since the reflecting surface of the beam splitter BS is the surface on the lower right, the light ray starting from the source S and undergoing reflection at the mirror M_2 passes through the beam splitter three times, while the ray reflected at M_1 travels through BS only once.
- The optical path length through the glass plate depends on its index of refraction, which causes an optical path difference between the two beams.
- To compensate for this, a glass plate CP of the same thickness and index of refraction as that of BS is introduced between M_1 and BS.
- The recombined beams interfere and produce fringes at the screen E.
- The relative phase of the two beams determines whether the interference will be constructive or destructive.
- By adjusting the inclination of M_1 and M_2 , one can produce circular fringes, straight-line fringes, or curved fringes.

- From the screen, an observer sees M_2 directly and the virtual image M_1' of the mirror M_1 , formed by reflection in the beam splitter, as shown in Fig.
- This means that one of the interfering beams comes from M_2 and the other beam appears to come from the virtual image M_1' .
- If the two arms of the interferometer are equal in length, M_1' coincides with M_2 .
- If they do not coincide, let the distance between them be d , and consider a light ray from a point S .
- It will be reflected by both M_1' and M_2 , and the observer will see two virtual images, S_1 due to reflection at M_1' , and S_2 due to reflection at M_2 .

These virtual images will be separated by a distance $2d$.

If θ is the angle with which the observer looks into the system, the path difference between the two beams is $2d\cos\theta$.

When the light that comes from M_1 undergoes reflection at BS, a phase change of π occurs, which corresponds to a path difference of $\lambda/2$.



- Therefore, the total path difference between the two beams is,
$$\Delta = 2d \cos \theta + \frac{\lambda}{2}$$

- The condition for constructive interference is then,

$$\Delta = 2d \cos \theta + \frac{\lambda}{2} = m \lambda, \quad m = 0, 1, 2, \dots \quad (1)$$

- For a given mirror separation d , a given wavelength λ , and order m , the angle of inclination θ is a constant, and the fringes are circular
- They are called *fringes of equal inclination*, or *Haidinger fringes*.
- If M_1' coincides with M_2 , $d = 0$, and the path difference between the interfering beams will be $\lambda/2$.
- This corresponds to destructive interference, so the center of the field will be dark.

- If the images S_1 and S_2 from the two mirrors are exactly the same distance away, $d=0$ and there is no dependence on θ
- This means that only one fringe is visible, the zero order destructive interference fringe, where

$$\Delta_{\text{destr}} = \frac{\lambda}{2} = \left(m + \frac{1}{2}\right)\lambda, m = 0$$

and the observer sees a single, large, central dark spot with no surrounding rings.

- If one of the mirrors is moved through a distance $\lambda/4$, the path difference changes by $\lambda/2$ and a maximum is obtained.
- If the mirror is moved through another $\lambda/4$, a minimum is obtained; moving it by another $\lambda/4$, again a maximum is obtained and so on
- Because d is multiplied by $\cos\theta$, as d increases, new rings appear in the center, faster than the rings already present at the periphery disappear, and the field becomes more crowded with thinner rings toward the outside.
- If d decreases, the rings contract, become wider and more sparsely distributed, and disappear at the center.
- For destructive interference, the total path difference must be an integer number of wavelengths plus a half wavelength,

$$\Delta_{\text{destr}} = 2d \cos \theta + \frac{\lambda}{2} = \left(m + \frac{1}{2}\right)\lambda, \quad m = 0, 1, 2, \dots$$

Fabry-Perot Interferometer

- The Fabry interferometer (variable-gap interferometer) was produced in 1897 by the French physicists Charle Fabry.
- It consists of two highly reflective and strictly parallel plates called an etalon.
- Because of the high reflectivity of the plates of the etalon (a device consisting of two reflecting glass plates, employed for measuring small differences in the wavelength of light using the interference it produces), the successive multiple reflections of light waves diminish very slowly in intensity and form very narrow, sharp fringes.
- These may be used to reveal hyperfine structures in line spectra, to evaluate the widths of narrow spectral lines, and to redetermine the length of the standard meter.

...Fabry-Perot Interferometer

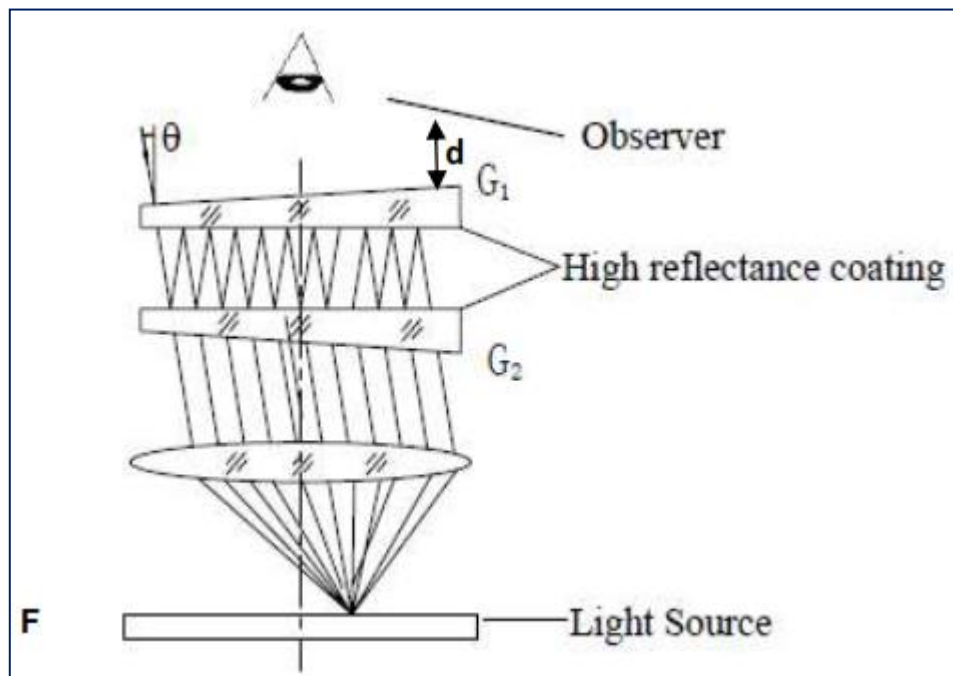
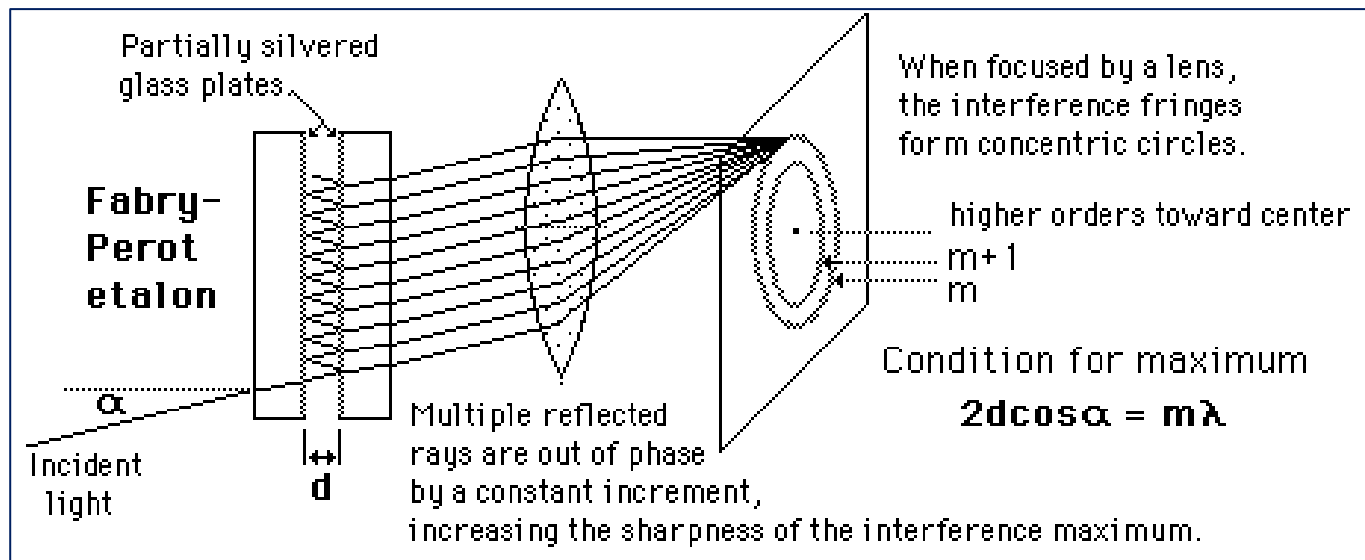
- The Fabry-Perot interferometer uses the phenomenon of multiple beam interference that arises when light shines through a cavity bounded by two reflective parallel surfaces.
- Each time the light encounters one of the surfaces, a portion of it is transmitted out, and the remaining part is reflected back.
- The net effect is to break a single beam into multiple beams which interfere with each other.
- If the additional optical path length of the reflected beam (due to multiple reflections) is an integral multiple of the light's wavelength, then the reflected beams will interfere constructively.
- More is the number of reflection inside the cavity, sharper is the interference maximum.
- Using Fabry-Perot (FP) interferometer as a spectroscopic tool, concepts free spectral range can be understood.

Principle of Working

- Two partial mirrors G1 and G2 are aligned parallel to one another at a distance d , forming a reflective cavity.
- When irradiated by a monochromatic light (a laser) of wavelength λ at an angle of incidence θ , multiple reflections take place inside the cavity.
- Part of the light is transmitted each time the light reaches the second reflecting surface.
- All such transmitted light rays interfere with each other to give rise to a maxima or minima depending on the path difference between them.
- Let n be the refractive index of the medium in the cavity (in this case it is air).
- Then the optical path difference difference between two neighbouring rays is :

$$\Delta = 2nd \cos \theta \quad \dots \quad \dots \quad (1)$$

Then the phase difference is given by $\delta = \left(\frac{2\pi}{\lambda}\right) \Delta \quad \dots \quad \dots \quad (2)$



Determination of wavelength λ

- Using the relations given below, wavelength of the incident light can be determined accurately.
- Let the initial separation between the mirrors is d_1
- If one counts the number of fringes (say maxima) appearing or disappearing at the centre ($\theta \sim 0$) by varying the distance between the mirrors to d_2
- then λ can be determined as follows:

$$\Delta = 2nd \cos \theta \quad \dots \quad (1)$$

$$I_T \text{ is maximum when } \Delta = m\lambda \ (m = 0, 1, 2, \dots) \text{ or } \delta = 2m\pi \dots \quad (4)$$

$$\text{and minimum when } \Delta = (2m+1)\lambda/2 \ (m = 0, 1, 2, \dots) \text{ or } \delta = (2m+1)\pi \dots \quad (5)$$

$$2d_1 = m_1 \lambda, \quad 2d_2 = m_2 \lambda, \quad m_2 - m_1 = \text{Number of maxima counted}$$

$$\lambda = \frac{2(d_2 - d_1)}{m_2 - m_1} \quad \dots \quad (6)$$